Algorithmic Discrimination in Car Insurance

Adriano Machado, André Silva, Cristiano Rocha, Francisco da Ana, Gonçalo Santos, and João Lima
Faculty of Sciences & Faculty of Engineering, University of Porto, Portugal
(Dated: October 25, 2026)

Michigan's 2019 car insurance reform intended to eliminate discriminatory pricing by banning non-driving factors from being used to calculate insurance premiums. However, insurance companies found loopholes in the law that allowed them to perpetuate racial and economic disparities through proxy variables and detailed geographic segmentation. Despite a sharp reduction in statewide rates, Detroit remained the most expensive city for car insurance, with Black neighborhoods still facing disproportionately high premiums. The case reveals the challenges of addressing algorithmic bias in insurance pricing, emphasizing the critical importance of transparency, explainability and fairness.

I. CASE STUDY ANALYSIS

A. Context

In Michigan, and particularly in the city of Detroit, car insurance has historically been an example of systemic economic inequality. For decades, the city held the distinction of having the most expensive car insurance rates in the United States.

Michigan's high car insurance costs were largely driven by its "no-fault" insurance system, which required every driver's insurance to cover their own damages and medical expenses, regardless of who caused the accident. A unique feature of this system was the mandatory unlimited Personal Injury Protection (PIP) coverage, which guaranteed lifetime medical benefits for accident victims. While this provided extensive protection, it made premiums much more expensive because insurers had to account for potentially enormous long-term costs. This combination of no-fault coverage and unlimited benefits meant that Michigan drivers, particularly those in Detroit, faced some of the highest insurance rates in the country.

In addition, insurance companies designed algorithms that used multiple non-driving factors to determine insurance rates. These factors included zip codes, sex, marital status, homeownership, educational level, occupation, and credit scores. As illustrated in Fig 1, this has led to disproportionately high car insurance costs for residents in the Detroit metropolitan area, particularly in the city itself and surrounding urban neighborhoods. It specifically penalized predominantly black populations, effectively creating a discriminatory pricing structure that went beyond individual driving performance.

The result was a systematic approach that consistently penalized residents in urban neighborhoods, particularly those with predominantly black populations, effectively creating a discriminatory pricing structure that went far beyond individual driving performance.

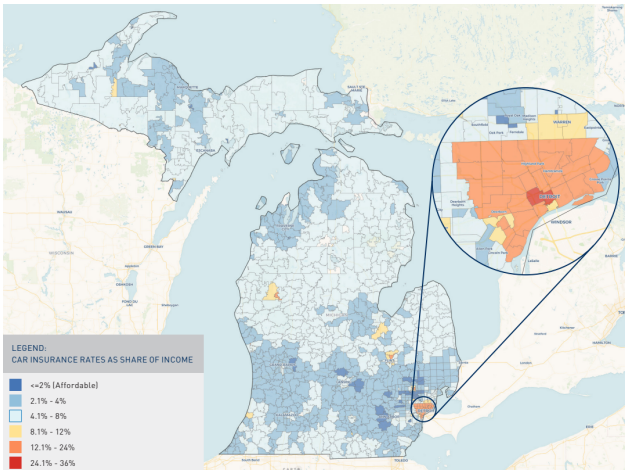


FIG. 1: Car insurance as percentage of pre-tax income by zip code as of 2018

In order to address the the extremely high car insurance prices in Michigan and, more importantly, reduce discriminatory practices, in 2019 the state of Michigan passed a legislative reform prohibiting the use of non-driving factors in calculating car insurance rates.

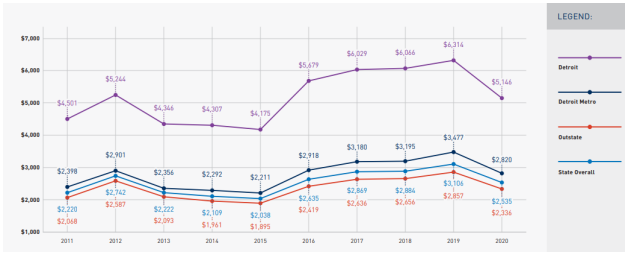


FIG. 2: Auto insurance rates, 2011 to 2020

After the legislation went into effect there was a significant reduction in insurance costs from 2019 to 2020 as we can see in Fig 2. However, Detroit remained the most expensive city and Michigan the most expensive state for car insurance.

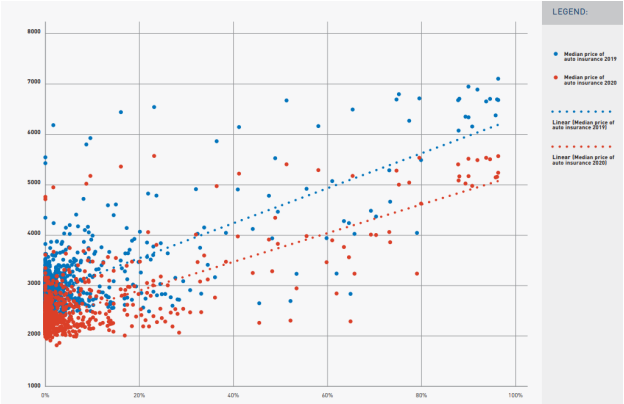


FIG. 3: Median price of auto insurance in Michigan zip codes 2019 and 2020 by percentage black in zip code

In terms of discrimination as the data in Figure 3 shows there is still a significant correlation between the percentage of black residents in a zip code and the median price of auto insurance. While the relationship has weakened from 2019 to 2020 (blue vs red regression line), the trend lines still indicate that zip codes with higher black populations continue to face disproportionately higher car insurance costs.

Key stakeholders included the insurance companies (Progressive, State Farm, Auto Club, Auto-Owners, Allstate, Citizens, Liberty Mutual), Michigan's regulatory body (DIFS - Department of Insurance and Financial Services), affected residents (particularly in Detroit), politicians, and consumer advocacy groups.

B. System details

A defining characteristic of insurance companies' pricing systems has been their refusal to disclose the details of their algorithmic pricing methodologies. These companies have long treated their pricing algorithms as secret, creating a black box that prevents independent verification of fairness and non-discrimination. When pressed about their methodologies, insurers provide only vague explanations, citing complex risk assessment models while revealing nothing substantive about their decision-making processes.

In order to investigate Michigan's car insurance practices, data journalists used a reverse engineering methodology. The team analyzed public insurance rate filings submitted to Michigan regulators and then cross-referenced these variables with demographic data, particularly focusing on race and income levels, to identify correlations between these factors and higher insurance premiums. This approach allowed the investigators to reveal hidden biases within algorithms and rating systems that were presented as impartial.

After the 2019 reform, insurers deployed new algorithmic systems that, while technically compliant with the law, maintained discriminatory pricing patterns through careful feature engineering and finer geographic segmentation. A key technical strategy was the development of proxy variables to replace banned features - for instance, "insurance scores" were derived from similar financial data as the prohibited credit scores, effectively preserving the same discriminatory patterns. Similarly, companies replaced zip code-based pricing with proprietary territory mapping systems. Allstate, for example, implemented a grid system where 84% of territory borders still aligned with zip code boundaries, as we can see in Fig 4.

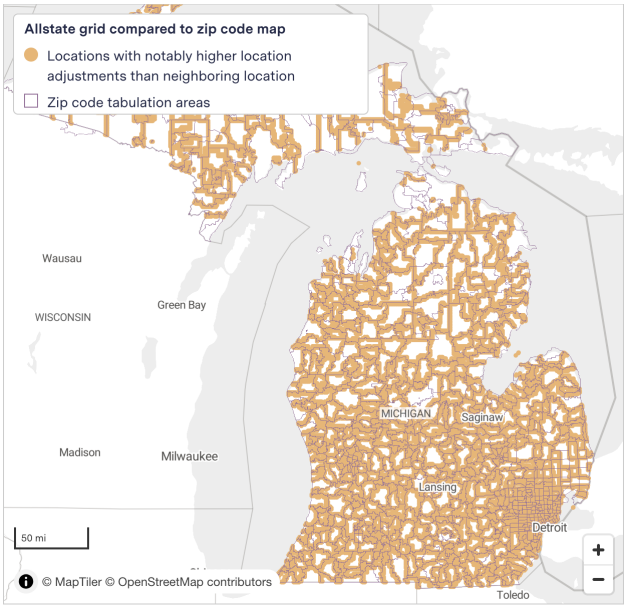


FIG. 4: Allstate location grid compared to zip code map

This strategy was allowed because the law states that, despite the restriction on postal zones, "Automobile insurance risks may be grouped by territory." Essentially, these territories could not be literal zip codes but could take on any other shapes. This led insurers to develop their own location systems, resulting in an even more rigorous and detailed classification of territories.

State Farm adopted a strategy similar to Allstate's. Auto Club chose a geographical division based on census tracts, which are small statistical subdivisions of a county used by the Census Bureau to analyze populations and housing characteristics. Liberty Mutual was the only insurer among these to switch to larger and broader geographical divisions: counties.

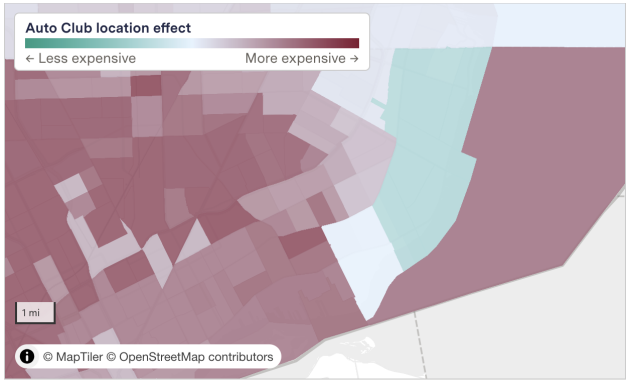


FIG. 5: Grosse Pointe and eastern end of Detroit under Auto Club pricing system

As we can see in Fig. 5, Auto Club's location-based system leads to highly questionable premium values. Some predominantly Black neighborhoods, such as Jefferson Chalmers and Morningside, are shown in shades of red on the map, indicating very high prices compared to a neighboring area called Grosse Pointe, represented in shades of blue. It is important to note that Grosse Pointe is a predominantly white neighborhood located less than a five-minute drive from the other mentioned areas.

The Markup's investigation into Michigan's car insurance practices exposed how insurers systematically circumvented reforms aimed at eliminating discriminatory pricing. These findings prove the persistence of systemic inequities, illustrating how technological opacity and regulatory loopholes can deviate well-intentioned reforms.

C. Positive and negative outcomes

The new algorithmic pricing systems led to mixed results, bringing both positive changes and persistent challenges. On the positive side, the reform reduced car insurance rates and introduced flexibility by removing the unlimited no-fault Personal Injury Protection (PIP) requirement, allowing policyholders to choose from multiple benefit levels. The previous mandatory acquisition of full coverage insurance contributed to the state's high insurance costs. The reform allowed drivers to choose lower coverage levels, potentially reducing premiums.

However, despite these efforts, Michigan remained the most expensive state of the country for auto insurance, with Detroit continuing to lead as the costliest city. While the average cost in Michigan decreased after the reform, even as the national average increased, it failed to address racial disparities effectively. Insurers like Allstate and Liberty Mutual saw a slight increase in the percentage of Black residents living in the most expensive areas post-reform, as algorithmic pricing systems did not eliminate racial biases in insurance premiums.

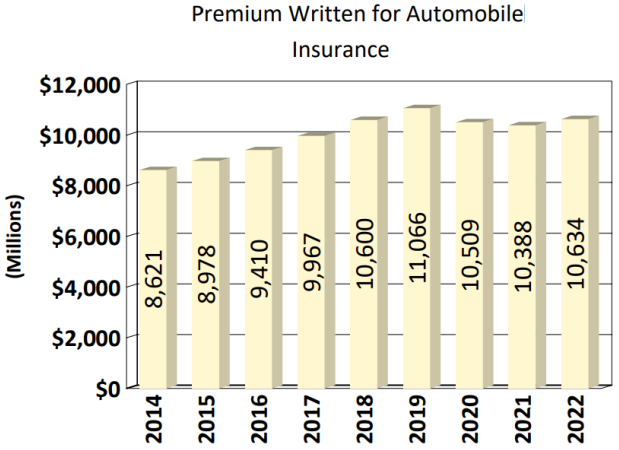


FIG. 6: Michigan Premium Written for Automobile - National Association of Insurance Commissioners (NAIC)

Fig. 6 shows the trend in premium written (total amount of insurance premiums collected by auto insurers) in Michigan from 2014 to 2022. After a continuous increase from 2014 to 2018, the premium written decreased following the 2019 reform, but remained high.

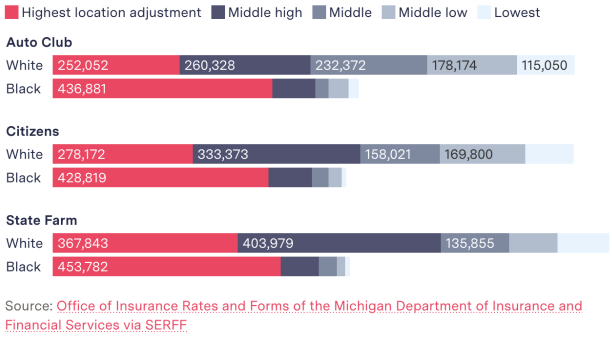


FIG. 7: Location-based premium adjustments for White and Black Michiganders

Fig. 7 evidences that Black residents disproportionately experienced the highest location-related adjustments. Insurers state that population density plays a large role in premium calculation: *"Urban areas have a greater population density, along with congested roads and highways ... Increased traffic and pedestrians on crowded roads lead to more crashes and therefore, more claims"*. However, if population density played such a central role in how much insurers charge, racial disparities should have disappeared within the state's most densely populated areas, since Michigan's densest areas have majority-White residents.

Moreover, the reform exposed critical flaws in the insurance industry's commitment to fairness, transparency, and explainability. Fairness is compromised as insurers continue to charge higher premiums in predominantly Black neighborhoods, even after the reform aimed to reduce discrimination. This indicates a failure to achieve

equitable outcomes for all communities. Transparency is lacking as the algorithms remain opaque, making it difficult for consumers and regulators to understand how rates are set and identify biases. Finally, explainability is insufficient, leaving consumers unaware of the factors that affect their premiums, reducing trust and accountability in the insurance system.

In conclusion, despite reducing costs and increasing coverage options, Michigan's auto insurance reform failed to eliminate racial disparities, highlighting ongoing challenges with fairness and transparency in algorithmic pricing.

D. Societal Response

1. Political Debate and Legislative Support

The legislative process highlighted significant ideological and regional divides. The Republican-led initiative aimed to reduce insurance costs by allowing consumers to opt out of mandatory unlimited Personal Injury Protection (PIP) coverage. This flexibility was promoted as a way to decrease premiums by offering drivers a choice in coverage levels.

However, opposition arose from within the Democratic Party, particularly the "Real Deal" caucus, which criticized the reform for potentially leaving accident victims underinsured and vulnerable to crippling medical bills. State Senator Stephanie Chang (D-Detroit) voiced concerns that reduced coverage options could disproportionately harm lower-income households, pushing them toward bankruptcy or reliance on Medicaid.

Despite these objections, the reform passed with bipartisan support: 89-20, with all Republicans and 32 Democrats voting in favor, while 20 Democrats voted against. This split underscored urban-rural divides within the party, reflecting differing priorities between Detroit, with its high concentration of Black residents facing systemic inequities, and other regions of the state of Michigan.

Since the reform, progress has been slow. Senate bills 530, 531, and 575, aimed at creating a reasonable fee schedule to address the care crisis, have stalled. Tom Judd, Executive Director of the Michigan Brain Injury Provider Council, stressed that these bills were crucial for improving care for accident victims. Despite support from the Governor and legislators, the bills remain blocked, leaving many without the care they need.

2. DIFS Oversight and AI Regulation

In August 2024, DIFS issued Bulletin 2024-20-INS: *Use of Artificial Intelligence Systems by Insurers*. This bulletin outlined guidelines for the use of AI systems by clarifying DIFS' expectations and outlining insurers' responsibilities. The aim is to prevent unfair discrimination, ensure compliance with existing laws and address concerns about transparency and data security. The bulletin emphasizes that while AI can bring innovation to the insurance sector, it also poses risks that must be mitigated, particularly those related to biased pricing outcomes, potential discrimination and lack of transparency.

Key requirements include creating a formal AI Systems (AIS) Program that properly document the development, implementation, and ongoing management of the system. The department may request detailed information about these systems during investigations, in order to ensure fairness and transparency of the predictive models. The ultimate goal is balancing technological innovation with consumer protection, ensuring that AI-driven decisions in insurance remain fair, transparent, and accountable.

II. TECHNICAL COMPONENT

After reviewing some key aspects that lead the Markup Magazine investigation, we started by looking for the dataset that Markup used for their investigation. Sadly, we didn't found any proper datasets that could be used in order to perform a strong analysis on this subject. Therefore we decided to look for other datasets from different cities within the same topic. After a long search in database banks such as **Kaggle**, **Google Dataset Search**, **UCI Machine Learning Repository**, among others, it was found the dataset from "Suckers List: How Allstate's Secret Auto Insurance Algorithm" on Kaggle (this dataset was used by the Markup to perform an analysis on the premiums that were being charged by Allstate's car insurance in the state of Maryland). This dataset provides detailed records of insurance premiums and related factors from citizens across Maryland counties where it included demographic information and premium data (prices from the insurance company). This offered us a unique opportunity to dissect the pricing mechanisms used by the insurance company and it allowed us to pursue our main goal: Understand how the different variables interplay in determining premiums and whether biases exist in pricing strategies.

A. What and why?

In our dataset, each record gives us the information about the zip code, gender, birth date, premiums (current, indicated, selected and underlying), the total expenses and the cgr factor of the correspondent citizen. These variables enabled us to perform a deeper investigation into potential disparities in pricing based on, for example, demographic factors. This was super important because our main goal was to develop an analysis to uncover patterns of algorithmic bias in insurance pricing specially, whether certain groups are systematically charged higher premiums despite similar risk profiles (identifying such biases was crucial for promoting

fairness and transparency in the insurance company). Besides that, we also needed to perform a little bit of research in order to fully understand the meaning of the different variables. The variables that needed our attention were: current premium, selected premium, indicated premium, underlying premium and total expanses. So, after looking for this information, we found our that these corresponded, respectively, to the premium that were being charged at the time, the premium that was going to be charged during a transition phase before updating to the final premium (the final value calculated by the insurance company, indicated premium). We also noticed that there was no information about the underlying premium and the total expanses variables.

B. How?

For our full analysis, we divided our process into these fundamental steps:

- **Data Profiling and Preparation:** The analysis began with the data overview and a data profile report to understand the distributions, correlations and potential anomalies within the dataset. In this phase, we also came across with some variables that we didn't think it should be included in our analysis because we didn't find any information to explain them as we have already explained in the subsection before. Besides that, we also added two variables into our data frame where the main goal was to divide the individuals into different categories. These categories would indicate if a certain person was subject to a increase or a decrease in the transition phase or after this phase (two variables: one with the total difference and another with the transition difference).
- **Fairness Analysis:** After performing the profiling and the simple preparation of our dataset, we went on to check for potential bias between the indicated premium and the territory. For this analysis, we mainly focused on using a specific measure of fairness, the disparate impact ratio to compare each variable with the indicated premium.
- **Predictive Modeling:** A linear regression model was built to examine the importance of the selected and current premium over the indicated premium and the importance of current and indicated premium over the selected premium.
- **Analyze the categories distribution:** After building the linear regression models, we started identifying outliers or anomalous pricing patterns in transition prices(calculated the residuals (the difference between actual observed transition prices and the predicted prices from a linear regression model) to assess the model's performance and highlight large discrepancies). Besides that, we also analyzed the overall distribution of the different categories, including the outliers ("Decrease or no change", "Low Increase", "High Increase"), and the respective percentage of individuals on each category.
- **Analysis on the demographic factors:** In this step, we mainly focused on the age, gender and territory variables. We performed an analysis on the gender distributions among the different categories, we visualized the distribution of men and women among the different age bins, we analyzed the age distribution with the different categories outcomes, the percentage of different categories outcomes by each age group and, at the end, we performed a full analysis on the price distribution between counties with higher percentages of non-white people. For this final analysis I enumerated, we needed to get extra information from the census in order to get all the information of each territory and county where it included the percentage of non-white people.
- **SHAP values analysis:** To end our analysis, we also performed a final step to try to fully understand the contribution of individual features to premium predictions. For that, Shapley values were computed and these provided a robust framework for attributing the variance in premium pricing to specific features, ensuring a transparent and explainable analysis of the pricing model.

C. Results and Discussion

Let's go through the main graphics of our analysis:

By analyzing the disparate impact measure by territory, we could clearly see that there were some potential bias for some specific territories, even though the chi-square test didn't give us many insights about the territory feature

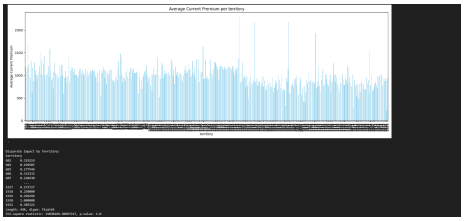
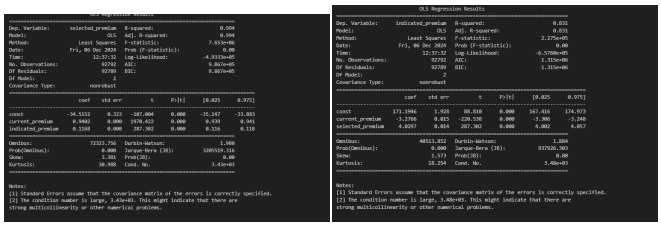


FIG. 8: Fairness between territory and indicated premium

On the left side of Figure 9, the regression analysis reveals that the dependent variable, selected_premium, is strongly explained by current_premium and indicated_premium, with an R-squared of 0.994, indicating that 99.4% of the variance is accounted for by the

model. Both predictors are highly statistically significant (p ¡ 0.001), with current_premium having a larger positive effect (coefficient = 0.9402) compared to indicated_premium (coefficient = 0.1168). Despite the model's strong explanatory power, diagnostics raise concerns about residual normality, as indicated by the significant Omnibus and Jarque-Bera tests, and potential multicollinearity, evidenced by a high condition number (3.43e+03). These issues warrant further investigation to ensure the robustness of the results.

In the right side of Figure 9, the regression analysis highlights that the indicated_premium is strongly influenced by both current_premium and selected_premium, with the model explaining 83.1% of the variance (R-squared = 0.831). Specifically, the current_premium has a negative effect, decreasing the indicated_premium by 3.28 units per unit increase, while the selected_premium has a positive effect, increasing the indicated_premium by 4.03 units per unit increase. Besides that, both predictors are highly statistically significant (p ¡ 0.001). However, diagnostic tests indicate that the residuals deviate from normality and suggest potential multicollinearity issues.



OLS Regression Results

Dep. Variable:	selected_premium	R-squared:	0.994
Model:	OLS	Adj. R-squared:	0.994
Method:	Least Squares	F-statistic:	7.453e+06
Date:	Fri, 06 Dec 2024	Prob (F-statistic):	0.00
Time:	12:37:32	Log-Likelihood:	-4.9333e+05
No. Observations:	92792	AIC:	9.867e+05
Df Residuals:	92789	BIC:	9.867e+05
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-34.5152	0.323	-107.004	0.000	-35.147	-33.883
current_premium	0.9402	0.000	1970.422	0.000	0.939	0.941
indicated_premium	0.1168	0.000	297.302	0.000	0.116	0.118

Omnibus:	72323.756	Durbin-Watson:	1.988
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3205519.316
Skew:	1.381	Prob(JB):	0.00
Kurtosis:	30.988	Cond. No.	3.43e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

OLS Regression Results

Dep. Variable:	indicated_premium	R-squared:	0.831
Model:	OLS	Adj. R-squared:	0.831
Method:	Least Squares	F-statistic:	2.275e+05
Date:	Fri, 06 Dec 2024	Prob (F-statistic):	0.00
Time:	12:37:32	Log-Likelihood:	-6.5700e+05
No. Observations:	92792	AIC:	1.315e+06
Df Residuals:	92789	BIC:	1.315e+06
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	171.1946	1.920	88.810	0.000	167.416	174.973
current_premium	-3.2766	0.015	-220.536	0.000	-3.306	-3.248
selected_premium	4.0297	0.014	287.302	0.000	4.002	4.057

Omnibus:	40513.852	Durbin-Watson:	1.804
Prob(Omnibus):	0.000	Jarque-Bera (JB):	937926.303
Skew:	1.573	Prob(JB):	0.00
Kurtosis:	18.254	Cond. No.	3.40e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.4e+03. This might indicate that there are strong multicollinearity or other numerical problems.

FIG. 9: Regression Models

Relatively to the categories distribution, the graphics: reveal that customers with price changes exceeding 9.5% (marked in red) consistently display higher residuals, indicating that the model struggles to predict their transition prices accurately; demonstrate the relationship between the current premium (x-axis) and the indicated premium minus the current premium (y-axis), grouped by different pricing change categories: "Low Increase" (blue), "High Increase" (red), and "Decrease or No Change" (gray). Customers experiencing significant price increases (red cluster) are concentrated at higher premium values, indicating that large increases are more likely at the top end of the premium spectrum; individuals in the "High Increase" category not only have higher premiums on average but also show a broader range of premiums; those in the "Decrease or No Change" category generally have lower premiums and less variability, indicating more uniformity in this group; across all categories, outliers exist, with "Decrease or No Change" having the most noticeable extreme values. This fact can clearly show that the model that is being used by the insurance company is imprecise and unfair.

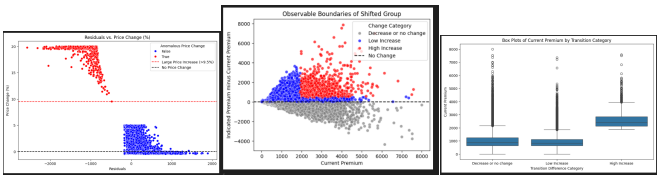


FIG. 10: Categories Distribution

By analyzing this graphics we can observe: **Gender-Based Trends:** across most age bins, males constitute a higher percentage in categories involving "Decrease or no change" and "High Increase," whereas females are more evenly distributed in the "Low Increase" category. In a lot of cases it is very different compared to the overall distribution like the high increase in the age bin of [20,30] and [30,40], for example; Notably, in the high increase category for older age bins (e.g., 60–70 and 70–80), males dominate with percentages exceeding 70%; **Age Group Patterns:** Younger age bins, [10–20] for example, show NaN or very limited instances of "High Increase" premiums, with no occurrences in this category. Conversely, older age groups (40–70) exhibit higher counts in all transition categories; The "Decrease or no change" category consistently represents the largest group across all age bins, indicating that most customers experience no significant price adjustments. Besides that, these scenarios exhibit higher counts in the age group of [30,70]. **Low Increase as a Common Outcome:** The "Low Increase" category sees the most balanced gender representation, especially in the 20–30 or 80–90 age bins, where male and female proportions approach parity.

Besides that, we can also take conclusions on the age distribution: **Categories Age Distributions:** "Low Increase" shows a slightly higher density among middle-aged individuals; "High Increase" peaks prominently around the age of 40 and 50, showing a strong concentration in that age range; "Decrease or no change" shows a more uniform spread across the ages but slightly higher densities among younger and older individuals; **Extra possible theories:** age might be a key factor or at least an important one since the category distributions vary significantly by age; The "High Increase" group being concentrated around middle-aged individuals could indicate that this age demographic is more responsive or susceptible to injustices in their insurance's prices.

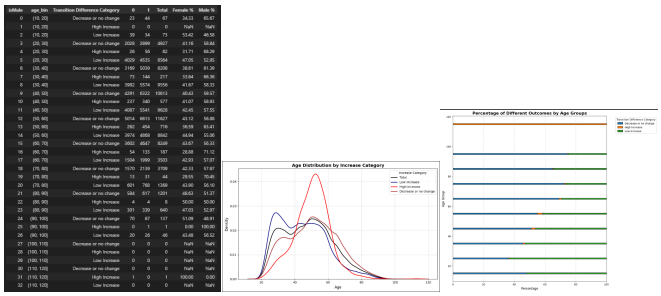


FIG. 11: Gender and Age analysis on the different categories

Relatively to the bucket graphic, it presents data on the proportion of price change categories (Decrease or no change, High Increase, and Low Increase) across different buckets of nonwhite population percentages. From the table, we observe that as the nonwhite population percentage increases from 0–25% to 75–100%, the proportion of areas experiencing "Decrease or no change" in price changes tends to increase. Specifically, for the 0–25% bucket, this proportion is 51.4%, while for the 75–100% bucket, it rises to 56.6%. Similarly, the category "Low Increase" shows a decreasing trend as the nonwhite population percentage rises, with proportions decreasing from 46.9% in the 0–25% bucket to 40.3% in the 75–100% bucket. Conversely, the "High Increase" category remains relatively low across all buckets, with proportions ranging from 1.8% to 2.7%. The bar chart confirms these trends visually. It shows that the blue sections (Decrease or no change) increase in size as the nonwhite population bucket increases, while the red sections (Low Increase) become smaller. The gray sections (High Increase) remain relatively consistent across all buckets, indicating a minor role in overall price changes.

In summary, areas with higher nonwhite populations tend to experience fewer "Low Increase" transitions and more "Decrease or no change" transitions in price changes, suggesting a relationship between demographic composition and pricing trends. However, the small and stable proportions of "High Increase" across all buckets indicate that significant price increases are less influenced by demographic factors in this data set.

On the other hand, the evolution of the median current premium over the county by an ascending Black or African American Population give an important insight about the correlation between the median current premium over the counties with an higher percentage of Black or African American citizens. In fact, this points also for a possible bias of higher insurance rates over populations that have a higher percentage of these citizens, given that this relation grows latently, that is, the most Black or African American citizens on a population there is, the more the current premium , in its median, increases, which holds true for every county of the State of Maryland.

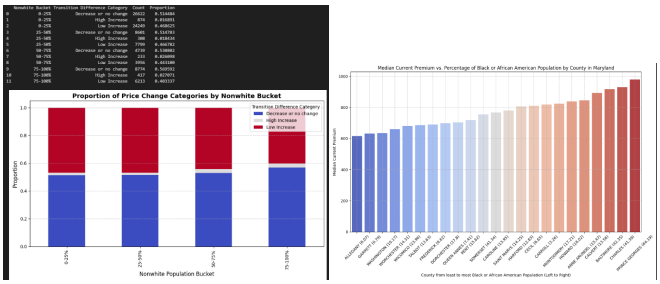


FIG. 12: Racial Analysis

However, while conducting a SHAP analysis, no direct correlation was found by either gender , age , or race. So, because of that, we decided to not include in our report since it adds no important insights.

III. CONCLUSION

Michigan's car insurance reform, while a step forward in addressing affordability, illustrates the complexities of dismantling systemic inequities. It serves as a case study in balancing regulatory intentions with real-world outcomes, emphasizing the need for practical and enforceable solutions to algorithmic bias.

The technical analysis of the "Suckers List: How Allstate's Secret Auto Insurance Algorithm" dataset reinforced these challenges. By utilizing techniques such as fairness metrics, linear regression, and Shapley values, the study uncovered significant disparities in premium allocation driven by geographic and demographic proxy variables. These insights demonstrated how algorithmic pricing models, even when technically compliant with regulations, can perpetuate systemic inequities through feature engineering and opaque methodologies.

This case highlights the challenges of regulating algorithmic pricing and the need for stronger transparency, fairness, and accountability in how insurance rates are set. To truly combat discrimination, regulators need to enforce stronger oversight and ensure that AI systems in insurance are used responsibly.

REFERENCES

[1] Julia Angwin et al. *Minority Neighborhoods Pay Higher Car Insurance Premiums Than White Areas With the Same Risk*. Published on April 5, 2017. ProPublica. Apr. 5, 2017. URL: https://www.propublica.org/article/minority-neighborhoods-higher-car-insurance-premiums-white-areas-same-risk (visited on 12/04/2024).

[2] Curbed Detroit. *Michigan Auto Insurance Reform Bill Explained*. May 2019. URL: https://detroit.curbed.com/2019/5/14/18623153/michigan-auto-insurance-reform-bill-explained.

[3] Poverty Solutions. *Auto Insurance and Economic Mobility in Michigan*. May 2019. URL: https://poverty.umich.edu/files/2019/05/auto_insurance_and_economic_mobility_in_michigan_2.pdf.

[4] Maddy Varner and Aaron Sankin. *Suckers List: How Allstate's Secret Auto Insurance Algorithm Squeezes Big Spenders. Insurers are supposed to price based on risk, but Allstate's algorithm put a thumb on the scale*. Published by The Markup. Feb. 25, 2020. URL: https://themarkup.org/allstate-algorithm/2020/02/25/suckers-list-how-allstates-secret-auto-insurance-algorithm-squeezes-big-spenders (visited on 12/04/2024).

[5] Maddy Varner et al. *Show Your Work: How We Analyzed Allstate's Car Insurance Algorithm*. Published by The Markup and Consumer Reports. Feb. 25, 2020. URL: https://themarkup.org/allstates-algorithm/2020/02/25/show-your-work-car-insurance-suckers-list (visited on 12/04/2024).

[6] Poverty Solutions. *Policy Brief: Auto Insurance Reform in Michigan*. Dec. 2021. URL: https://sites.fordschool.umich.edu/poverty2021/files/2021/12/PovertySolutions-Auto-Insurance-Reform-PolicyBrief-December2021.pdf.

[7] Tracy Samilton. *Reports: Michigan's 2019 no-fault overhaul is failing to lower rates and help Detroit drivers as promised*. Published on May 4, 2022. Michigan Public. May 4, 2022. URL: https://www.michiganpublic.org/politics-government/2022-05-04/reports-michigans-2019-no-fault-overhaul-is-failing-to-lower-rates-and-help-detroit-drivers-as-promised (visited on 12/04/2024).

[8] Insurance Research Council (IRC). *Auto Insurance Reforms in Michigan Helping to Improve Affordability*. 2024. URL: https://www.insurancebusinessmag.com/us/news/auto-motor/auto-insurance-reforms-in-michigan-helping-to-improve-affordability--irc-500496.aspx.

[9] Mohamed Al Elew and Aaron Sankin. *How We Investigated Car Insurance Loopholes in Michigan*. Published on July 18, 2024, 06:15 ET. The Markup. July 18, 2024. URL: https://themarkup.org/premium-penalty/2024/07/18/how-we-investigated-car-insurance-loopholes-in-michigan (visited on 12/04/2024).

[10] Department of Insurance and Financial Services (DIFS). *DIFS Issues Bulletin to Address Use of AI in the Insurance Industry*. Issued on: August 7, 2024. State of Michigan Department of Insurance and Financial Services. Aug. 7, 2024. URL: https://www.michigan.gov/difs/news-and-outreach/press-releases/2024/08/07/difs-issues-bulletin-to-address-use-of-ai-in-the-insurance-industry (visited on 12/04/2024).

[11] Elle Meyers. *Advocates continue to call for changes to Michigan no-fault auto insurance*. Updated on: May 29, 2024 / 6:11 PM EDT / CBS Detroit. CBS News Detroit. May 29, 2024. URL: https://www.cbsnews.com/detroit/news/michigan-no-fault-auto-insurance-reforms/ (visited on 12/04/2024).

[12] National Association of Insurance Commissioners (NAIC). *Key Facts and Market Trends: Michigan Auto Insurance*. 2024. URL: https://content.naic.org/sites/default/files/publications-key-facts-market-trends-michigan.pdf.

[13] Aaron Sankin et al. *Michigan's "Fair and Reasonable" Reforms Allowed Car Insurers to Charge More in Black Neighborhoods*. An investigation by The Markup and Outlier Media. July 2024. URL: https://themarkup.org/premium-penalty/2024/07/18/michigans-fair-and-reasonable-reforms-allowed-car-insurers-to-charge-more-in-black-neighborhoods.